



Web Design and Development Assignment 1

HND in Computing (ESOFT Metro Campus)



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WEB DESIGN AND DEVELOPMENT

Unit 10: Assignment 1(Online Hospital Management System)



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Task 1

1.1

Server technologies and management services associated with hosting and managing websites

Explanation of web technologies:

Web technology is the creation and application of systems that enable resource sharing and communication between various computers and gadgets. Web technologies are the foundational elements of every functional computer network. Web technologies are infrastructural building blocks of any effective computer network: local area network, metropolitan area network or a wide area network, such as the Internet. Communication on a computer could never be as effective as they are without the plethora of Web technologies in existence. (Naveenkumar, medium.com, 2020)

Use of web technologies

- Computers and other network devices need to communicate.
- A mechanism must make it possible for a computer to communicate with another computer on the same network or another network.
- The transmission of a message from the sender to the recipient must be ensured by the mechanism, allowing the recipient to recover the message, respond, and confirm receipt or failure of communication.
- A variety of Web technology is vital to the function and success of many businesses.
- These include of websites, apps for arranging appointments online, and channels for client interaction. Additionally, Web technology enables companies to gather user data to further customize their offerings. (Naveenkumar, medium.com, 2020)

How web technologies are developed?

1. By using Markup Languages:
 - Markup languages like HTML, CSS, and XML are part of Web technology.

- These languages tell computers in text how to format, layout and style Web pages and programs.
- Two types of markup languages include procedural markup and descriptive markup. Additional types of languages include CGI and HTTP. (Naveenkumar, medium.com, 2020)

2. By using Programming Languages:

Perl, C#, Java, and Visual Basic.NET are some examples of programming languages. Web developers use these languages to make websites and applications. Each language has advantages and disadvantages, and most developers are familiar with a variety of languages to aid in their objectives.

For example:

HTML: The Foundation of any Web Site.

HTML (Hyper Text Mark-up Language) is the glue that holds together every web site. Like building a house, you always build a strong foundation first. For any site, HTML is that foundation. HTML is an open-source language (i.e. not owned by anyone), which is easy to learn, and requires no fancy (or expensive!) packages to start using it. All you need is something to type with, such as Windows Notepad, and a lot of time and patience. (Naveenkumar, medium.com, 2020)

3. By using stylish language:

Like

CSS

- CSS is a relatively new language, designed to expand upon the limited style properties of HTML.
- Easy to learn and implement, CSS is an excellent way to control the style of your site, such as text styles like size, color, and font.
- CSS may also be placed inside the HTML page or in separate files.

Java script

- JavaScript is an object orient programming language designed to make web development easier and more attractive. In most cases, JavaScript is used to create responsive, interactive elements for web pages, enhancing the user experience.

- Things like menus, animations, video players, interactive maps, and even simple in-browser games can be created quickly and easily with JavaScript.
- The real advantage of having all of the style properties for your entire site in one single CSS file is that you may edit that single file to effect changes on the whole site, rather than having to go through each HTML file one at a time. For this reason, it is perhaps the most useful web technology and certainly one of my favorites. (Naveenkumar, medium.com, 2020)

Different Web Technologies

- Browsers.
- HTML and CSS.
- Programming Languages.
- Frameworks.
- Web Servers.
- Databases.
- Protocols.
- Lastly

Communication protocols

Networked communications equipment must agree on a number of physical features of the data that is to be transmitted before a successful transmission can occur. "Protocols" are the set of guidelines that govern data communications.

In a largely unregulated digital environment, communications protocols create rules.

For example, on the Internet, the types of communications protocols created by groups like the World Wide Web Consortium or W3C and the Internet engineering task force or IETF help to provide universal operations and limit various kinds of liability and vulnerabilities in these technologies.

One good example is authentication protocols. The authentication protocols work against things like distributed denial of service (DDoS) attacks and other common methods by which errant hackers can hobble vast networks or compromise or confuse network traffic.

When HTTP becomes HTTPS or the widespread use of SSL certificates becomes the norm, this is what's intended by those establishing those types of communications protocols. The protocols work to "harden" systems against various kinds of abuse inherently, and that's a big part of their value.

The use of data packets in global network paths is governed by other communications protocols, which can occasionally resemble particle physics. It's not like an open faucet where information merely flows freely from one place to another.

The message is made up of distinct, granular units between the sender and the receiver, with each individual data packet having its own header, core data, and routing trajectory. Communications protocols play such a strong and crucial role in this because everything needs to be carefully coordinated and planned.

Additionally, the underlying technology itself affects the communication protocols that are employed. Consequently, protocols also advance along with technology. Because of this, Internet protocols will continue to improve and we'll probably witness very different communication protocols in tomorrow's developing networks.

As ever more sets of devices are connected to a global network, the Internet of Things is predicted to increase demand for new, changing communications protocols. (Techopedia, 2019)

Purpose of communication protocols

1. Line level compatibility

One of the challenges in data communications is that different pieces of equipment use different voltages to represent 1s and 0s. For example, some devices use -5 volts to represent a 1, while others use +5 volts. This can create potential problems when connecting devices together, as a -5 volt signal may be interpreted as a 0 by a device that is expecting a +5 volt signal.

To avoid these conflicts, devices must be designed to be line-level compatible, which means that they can translate between the various voltage levels, in order to communicate with other devices.

While this may seem like a relatively simple concept, it is one of the biggest headaches in ensuring smooth and reliable data communication. Without proper standards, it is almost impossible to meet. (Mohammad Afaneh, Novelbits, 2019)

2. Timing relationship

A second important aspect that protocols must take into account is the timing relationships between devices. The data level transitions are determined by a clock at the sender (source). The receiver usually has no access to this clock.

As a result, the relative timing of the signals has to be recovered/ regenerated at the receiver to accurately determine the beginning and the end of the message. Protocols must, therefore, define how the receiver can reconstruct the original clock signal from the data signal in order to correctly interpret the timing relationships. This process is called Clock Recovery. (Mohammad Afaneh, Novelbits, 2019)

3. Time Management of Multiple Users

Another big issue that protocols must contend with is the time management of multiple users. Remember, in a network, there can be many devices sending and receiving data at the same time.

The protocols have to ensure that all these devices have fair access to the medium through which they are communicating and that no individual user dominates the medium for an excessively long period of time. (Mohammad Afaneh, Novelbits, 2019)

4. Error Control or Effects of Noise on the line

A fourth important consideration is the effects of noise on the line. In any practical communication system, there will always be some background noise present on the line, which can potentially corrupt or destroy the data being transmitted.

The job of the protocols is to provide mechanisms to detect and/or correct these errors so that the data is received in its original form. (Mohammad Afaneh, Novelbits, 2019)

5. Security

With the increase in IT hacking, security has become a major concern for embedded systems. A protocol must have provisions for security so that the data being transmitted cannot be accessed by unauthorized persons. (Mohammad Afaneh, Novelbits, 2019)

6. Power Control

If everyone were to transmit at the power above the maximum required for a satisfactory level of performance, it would create a lot of interference in the shared RF spectrum, and no one would be able to communicate properly.

It is the task of a communication protocol that uses the RF spectrum to make sure that each transmitter only transmits at the power required to cover the desired distance and no more. This allows others to use the same or nearby frequencies without interference. (Mohammad Afaneh, Novelbits, 2019)

7. RF Band Planning (Channel Spacing & Bandwidth)

For a protocol that uses the RF spectrum (e.g. Bluetooth LE), two important things are considered: how much space there is between channels, and how much bandwidth each channel can use.

Every communication protocol must use the correct amount of channel spacing and bandwidth for the RF Frequency band it will use. If it does not, then the systems that rely on the protocol will not work properly and could cause interference for other users of the RF spectrum. (Mohammad Afaneh, Novelbits, 2019)

8. Message Formatting

Another thing that a protocol must define is the message format. This includes the message structure and the sequencing of messages. The message structure is the way in which the data is arranged within the message.

For example, a message may have a header followed by a payload and then a checksum. The sequencing of messages is the order in which they are sent. For example, one message may need to be sent before another message in order to establish a connection. Take a look at how Bluetooth LE structures its packets and how it sequences them. (Mohammad Afaneh, Novelbits, 2019)

9. Techniques to Combat Reliability Issues

Communication is said to be reliable if the information sent is information received. But since the mediums used are often not 100% reliable, techniques must be used to combat the unreliability.

In the case of Bluetooth LE, some techniques include using a spread spectrum, RSSI as a link quality estimator, and a good modulation scheme (GFSK), among others. (Mohammad Afaneh, Novelbits, 2019)

Server hardware

A hardware server means that it is not a virtual machine. It means that it is a physical server.

Another meaning could be that that particular machine has the special hardware designed for servers, more reliable than the hardware for regular computers, plus the extra monitoring tools that are useful for managing/maintaining/monitoring a server but totally useless (generally speaking) for a regular computer.

A server is a computer equipped with specific programs and/or hardware that enables it to offer services to other computers (clients) on its network. There are different types and capabilities of servers. Think about transportation. We can think of transportation as anything that can move something or someone from one location to the other. A bicycle can move one person, a car can move four people, a bus can move 50 people, and a plane can move 500 people. They are all modes of transport, but each has a different capacity. The same applies to servers.

The term "server" can refer to either hardware or the software that runs on that hardware, which makes it a little unclear most of the time. Here, we are referring to the specific hardware itself. Servers are computers built to respond to requests from users or other computers and take appropriate action. Their primary goal is to support you, communicate information, and carry out duties to keep your workflow efficient and productivity high.

There are numerous same essential elements found in server gear, including memory, capacity, and CPUs. However, the similarities end there because server components require a more intensive, dedicated function than a conventional desktop does, so they are more specialized. It supports physical data in charge with the web-connected devices listed below and is connected to the internet. (Biren Negi, Quora, 2020)

Purpose of Server hardware

A server is a computer connected to a network of other workstations called 'clients'. Client computers request information from the server over the network. Servers tend to have more storage, memory and processing power than a normal workstation. How a server functions depends on the type of network it is on. Let's dive into a few different types of networks and how servers interact with them.

Peer-to-peer: In a peer-to-peer network, each workstation plays the role of client and server. The server is the computer that is providing information or services to the other computer. The networks rely on each other to provide and share information and services. These are typically only used in small offices or homes.

Client / Server: In a client/server network there is one server that is only dedicated to provide services to the other client workstations. You can have multiple servers in these networks but they cannot function as workstations. Client/server networks are the most commonly used.

Specialized Servers: There are a few common types of specialized servers. These include DNS, Web or Mail servers. These can hold databases of internet domains, supply access to websites or to email accounts. (Promax, 2019)

1. Computer

The majority of modern computers and gadgets meet the specifications needed to be used for creating websites. The quantity of the storage space is the most crucial setting to check on the computer; it should be greater than 2 GB because more is simpler. This will ensure that the computer runs quickly and without difficulty, especially while using resource-intensive tools like website builders or photo editors. A mouse and keyboard should be connected to the computer and work.

2. Internet

The ability to control a fast Internet connection could be crucial for the development of a website. This enables you to upload modifications to your website, send the files that make up your website to the web server that will host it, and check your website's functionality in real-time online.

3. External Hard Drive or DVDs for Backup

You should always save a current backup of your website in case there is an issue, such as a site accident or corrupted files. You can retain this storage on a set of DVDs or an external hard drive, depending on your viewpoint. An external drive could crash and you would lose your storage, but DVDs might be more troublesome and could also lose data as a result. How often you

update your site will determine how much of a storage you need, however for a website that is updated frequently, it can be once a month.

4. Server

To make sure that everyone who tries to visit your website can, you would need a client computer to host it. It is exceedingly difficult to set up a laptop to function as a remote server, and it is far easier, more efficient, and more cost-effective to simply purchase a service package from a domain registration company. They will continue to keep the server computer, and you can access it to make changes to the files for your page.

5. Mobile Devices

Many free website-building tools allow users to edit, maintain, and manage their page on a mobile device even though they are not required to do so. Consider getting a mobile device or computer with internet access if you are constantly on the road and want to be able to make adjustments to your website from wherever you are. (Promax, 2019)

Operating systems

An Operating System is the low-level software that supports a computer's basic functions, such as scheduling tasks and controlling peripherals.

An Operating System (OS) is an interface between a computer user and computer hardware. An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers. An operating system is software that enables applications to interact with a computer's hardware. The software that contains the core components of the operating system is called the kernel.

The primary purposes of an Operating System are to enable applications (software's) to interact with a computer's hardware and to manage a system's hardware and software resources. Some popular Operating Systems include Linux Operating System, Windows Operating System, VMS, OS/400, AIX, z/OS, etc. Today, Operating systems is found almost in every device like mobile phones, personal computers, mainframe computers, automobiles, TV, Toys etc. (Hackrich Team, Tutorials point, 2019)

Ex: Windows, Mac OS, UNIX, Linux

Purpose of Operating systems

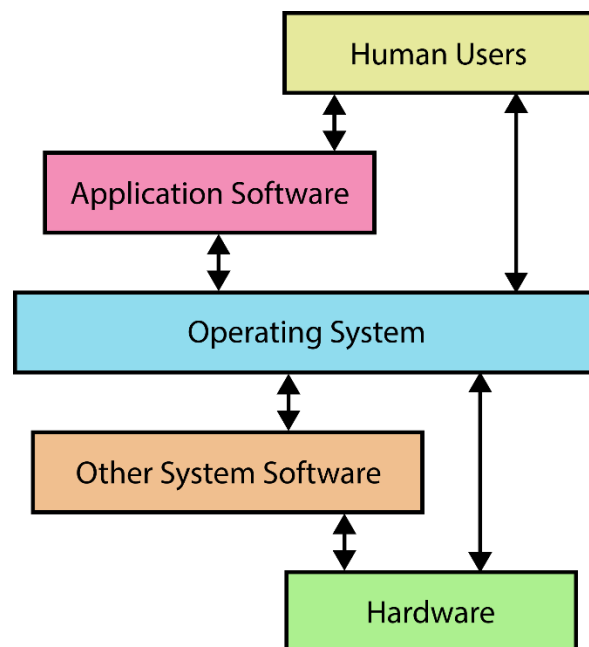
Most people use computers because they want to use application software (such as word processing, internet browsing) however these applications only work inside of

an operating system environment. The operating system is the software that controls, manages and run all of the applications on the computer.

The functions of an operating system include:

- Loading a software interface for the user to interact with (usually a GUI or CLI)
- Managing hardware resources such as memory and devices
- User management
- authentication & security
- File management

When a user is using a computer they are either interacting directly with the operating system (e.g. logging in to the computer) or via an application (e.g. saving a word processing document). (McMaster University, Learnlearn, 2021)



In an operating system software performs each of the function:

1. **Process management:** The OS can add and remove processes with the use of process management. Additionally, it offers systems for interprocess communication and synchronization.
2. **Memory management:** The memory management module manages the process of allocating and releasing memory to applications that require it.
3. **File management:** It controls all aspect of file management, including file organization, storage, retrieval, naming, sharing, and protection.
4. **Device Management:** All devices are tracked through device management. The I/O controller is a module that performs this function as well. It also manages the devices' allocation and de-allocation tasks.

5. **Security:** The security module guards a computer system's data and information against unauthorized access and malware threats.
6. **Command interpretation:** In order to process the commands, this module interprets the instructions given by the acting system resources. (McMaster University, Learnlearn, 2021)

Web Server Software

Web server is a computer where the web content is stored. Basically web server is used to host the web sites but there exists other web servers also such as gaming, storage, FTP, email etc.

The term web server can refer to hardware or software, or both of them working together.

On the hardware side, a web server is a computer that stores web server software and a website's component files (for example, HTML documents, images, CSS stylesheets, and JavaScript files). A web server connects to the Internet and facilitates physical data interchange with other devices connected to the web.

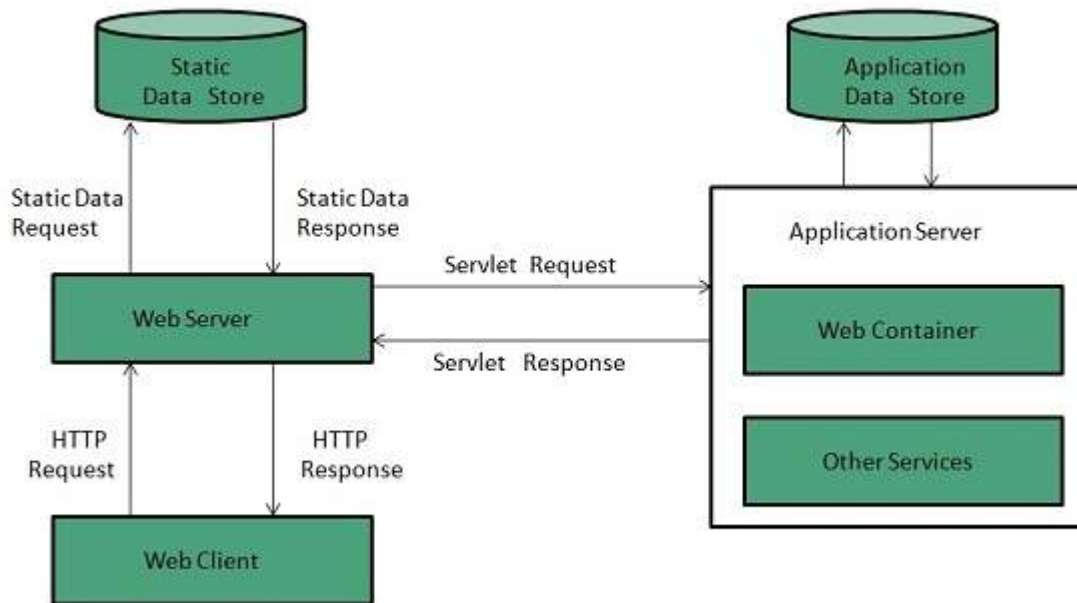
On the software side, a web server includes several parts that control how web user's access hosted files. At a minimum, this is an HTTP server. An HTTP server is software that understands URLs (web addresses) and HTTP (the protocol your browser uses to view webpages). An HTTP server can be accessed through the domain names of the websites it stores, and it delivers the content of these hosted websites to the end user's device. (MDN web docs, 2022)

Purpose of Web Server Software

Multi-processing: Several single-threaded child processes are started by the parent process, which also distributes incoming requests to the child processes. Each child process is in charge of dealing with a single request. The parent process is in charge of keeping an eye on the load and choosing whether to kill or fork processes.

Multi-threaded: Unlike Multi-process, it creates multiple single-threaded process.

Hybrid: It combines the two methods mentioned previously. This method creates numerous processes, each of which starts a number of threads. One connection is handled by each thread. Less strain is placed on system resources when numerous threads are used in a single process. (Tutorialspoint, 2018)



Examples

1. Apache HTTP Server

The Apache Software Foundation created the most widely used web server in the world. Almost every operating system, including Linux, UNIX, Windows, FreeBSD, Mac OS X, and more, can run the free source Apache web server. The Apache Web Server is used by about 60% of web server machines. (Tutorialspoint, 2018)

2. Internet Information Services (IIS)

The Internet Information Server (IIS) is a high performance Web Server from Microsoft. This web server runs on Windows NT/2000 and 2003 platforms (and may be on upcoming new Windows version also). IIS comes bundled with Windows NT/2000 and 2003; Because IIS is tightly integrated with the operating system so it is relatively easy to administer it. (Tutorialspoint, 2018)

3. Lighttpd

The FreeBSD operating system includes the free web server lighttpd, which is pronounced lighty. This free, open-source web server uses a lot less CPU power and is quick and safe. Additionally, Lighttpd is compatible with Windows, Mac OS X, Linux, and Solaris. (Tutorialspoint, 2018)

4. Sun Java System Web Server

This Sun Microsystems web server is appropriate for medium-sized and big websites. Despite being free, the server is not open source. But it works with UNIX, Linux, and Windows operating systems. JSP, Java Servlets, PHP, Perl, Python, Ruby on Rails, ASP, Coldfusion, and other Web 2.0-related languages and technologies are supported by the Sun Java System web server. (Tutorialspoint, 2018)

5. Jigsaw Server

The World Wide Web Consortium is the source of Jigsaw (W3C's Server). It is open source, free, and compatible with a number of operating systems, including Linux, UNIX, Windows, Mac OS X, and Free BSD. Jigsaw is a Java software that can execute both PHP and CGI scripts. (Tutorialspoint, 2018)

Relationship between Communication protocol, Server Hardware, Operating system and web server software.

An agreement between communicating devices is represented by a communication protocol, which is first and foremost a set of guidelines that permits two or more entities in a communication system to communicate information through any modification of a physical quantity. Similar to how two devices would be connected but unable to communicate without a protocol, a protocol specifies the structure of packets and the guidelines for sending them over a network. Additionally, different protocols offer various levels of service, including basic error correction, multimedia data retrieval, and others. Additionally, there are four parts to data communication: the message, the sender, the receiver, the transmission medium, and the protocol. Conventions can also describe a wide range of transmission characteristics.

Similarly to this, web server hardware facilitates the delivery of internet-accessible material and is important for web traffic since it must be ensured that HTTP requests are handled with the required minimal response time. In the customer's eyes, the web server hardware is also important, and it is regarded impolite for a service provider to make their client wait mostly online. Similar to this, web server computers use multiple processors, but only a small number of desktop PCs have more than one processor. Server hardware comes in a variety of forms, including web servers, mail servers, and file servers. Web servers are used to run the Apache HTTP server, which makes websites accessible via the internet.

Furthermore, webserver software manages Apache or Microsoft IIS, which offers online access to hosted webpages. Any computer can be used as a web server as long as it is connected to the Internet. A web server is a program that uses HTTP to serve the files from web pages to users. It also includes a physical server, a server operating system, and software that is used to facilitate HTTP communication. The majority of Web servers are connected to the Internet via a high-speed connection, offering OC-3 or faster data transmission rates.

Additionally, an operating system known as the host operating system is required to run the web server hardware, software, and communication protocols. The host operating system uses container-based virtualization, where containers are logical partitions that are used to separate applications on the same server and enable those applications to share the same operating system while maintaining hardware isolation. The key advantages of a host operating system are its ability to run different operating systems simultaneously on the same machine and the creation of a virtual hard disk,

which stores all of the data and does not interfere with the host operating system in the event of a virtual hard disk crash.

The communication protocol, web server hardware, software, and operating system are also necessary for publishing and accessing the website; if any of these are missing, we cannot publish or access the website. Similar to how publishing a website requires both hardware and software for the web server, sending a message to the admin also requires a communication protocol. Additionally, because hosting is used to publish the website, the host operating system is also necessary. Because they are interconnected, if one stops working, the website will not function or function incorrectly. Finally, each component is necessary in order to publish and visit the website. (Tutorialspoint, 2018)

1.2

Identify and define the types of DNS and the uses of it, with clarifications on how domain names are structured.

DNS

Definition of DNS

All computers on the Internet, from your smart phone or laptop to the servers that serve content for massive retail websites, find and communicate with one another by using numbers. These numbers are known as IP addresses. When you open a web browser and go to a website, you don't have to remember and enter a long number. Instead, you can enter a domain name like example.com and still end up in the right place.

A DNS service such as Amazon Route 53 is a globally distributed service that translates human readable names like **www.example.com** into the numeric IP addresses like 192.0.2.1 that computers use to connect to each other. The Internet's DNS system works much like a phone book by managing the mapping between names and numbers. DNS servers translate requests for names into IP addresses, controlling which server an end user will reach when they type a domain name into their web browser. (Amazon, 2021)

Purpose of DNS

The DNS will continue to associate any website's domain name with its IP address. The primary function of DNS is to map symbolic names to IP addresses in a globally distributed, hierarchical database. Names are grouped in a tree form to allow for delegation of duty, which means that if one server doesn't know the binding, it moves up the hierarchy. Similarly, it defines the DNS protocol, which describes the specifics of the data format and data communication exchanges utilized in the DNS, as well as the technical functioning of the database service.

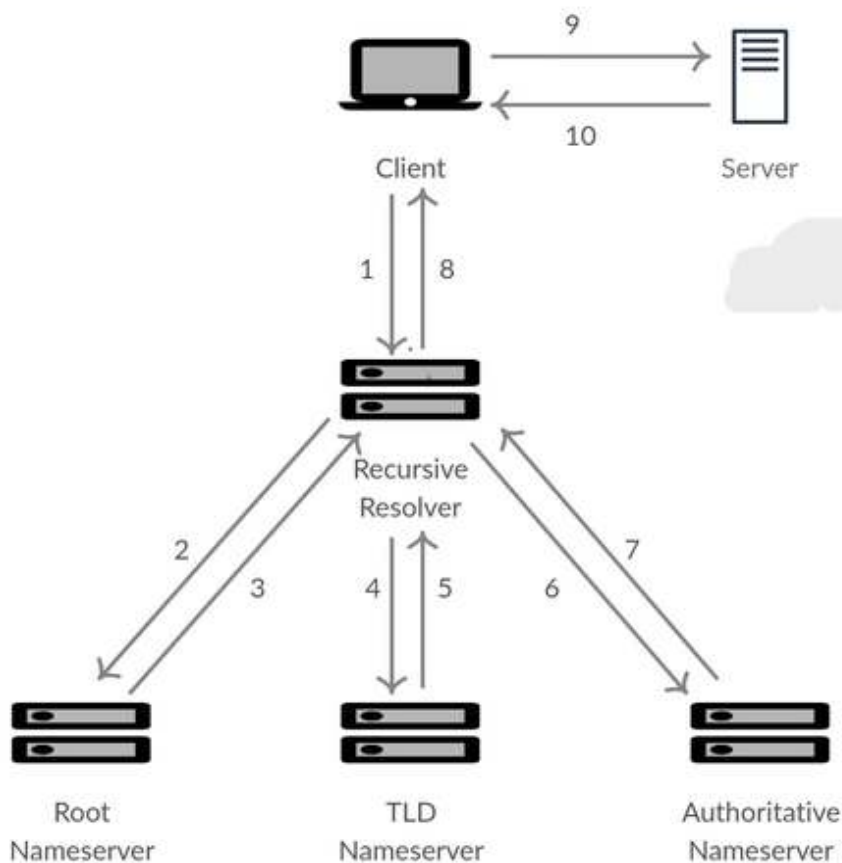
Additionally, hosting internet applications on servers close to your end users helps speed up domain name resolution, allowing users to access your website more quickly than other sites. Additionally, DNS enables the same naming convention as the internet, permits connections to local servers, and has DNS names that are more dependable than IP addresses. (Amazon, 2021)

Types of DNS

All DNS servers fall into one of four categories: Recursive resolvers, root nameservers, TLD nameservers, and authoritative nameservers. In a typical DNS lookup (when there is no caching in play), these four DNS servers work together in harmony to complete the task of delivering the IP address for a specified domain to the client (the client is usually a stub resolver - a simple resolver built into an operating system).

DNS servers convert domain names into IP addresses. It takes request from human and converts that into machine IP addresses that make easier to reach the origin server. (Morgan Joel, EDUCBA, 2022)

- **DNS Resolver:** DNS resolver is also called a recursive resolver, it converts hostname into IP address which is machine friendly. This server is designed to receive queries from web browsers and other applications. This resolver receives a hostname in the form of `www.sample.com` and also responsible for tracking the IP address for that hostname. (Morgan Joel, EDUCBA, 2022)
- **DNS Root Server:** Root servers are the DNS name servers that are operated in the root zone, root zone is the stored record of queries so that server can directly answer the query from the root zone or cache within the root zone. The DNS uses 13 DNS servers and that is based on a constraint of Internet Protocol version 4. (Morgan Joel, EDUCBA, 2022)
- **Authoritative Name Server:** The authoritative name server is the higher level in DNS. The DNS server is the 'authoritative', the authoritative name is given to the specific hostname because it holds the up-to-date information about that hostname. (Morgan Joel, EDUCBA, 2022)



Explanation on how domain names are organized and managed

Before the domain structure was created, computer names and IP addresses were managed using a flat list and stored in a plain text file known as a host file. Every domain is made up of two components: the actual domain and the top-level domain. Domain names were created to organize and find the IP address of computers on a large distribution network. Similar to this, ICANN has authority over and manages domains, and they are also contracted by ICANN to administer the domain for a specific period of time. Additionally, throughout history, different organizations have managed top-level domains. Among other requirements made public by ICANN in its written record agreement, organizations operating high-level domains are also in charge of the authoritative name server and the upkeep of a written record database. Additionally, these 13 servers maintained records on unique servers that were not highly valued statistically.

Additionally, domain management refers to the ongoing task of maintaining a personal or corporate domain that will be more stable, secure, and able to support related technologies. It also saves the information that may be specific to the workplaces that are current and complete database, arranges the domains browser that went to the customers, and after that it bypassed to the internet service provider DNS server. Additionally, the goals and needs of the domain owner must be taken into consideration while managing a domain because there are various tools available, but

only authentic tools should be used as they can be used to renew or cancel domain registrations. In addition, domain management tools and services can help to keep domain secure, stable and reliable.

The information to the left of the period and the letter combination to the right of it in a web address make up the domain structure, according to United Domains. The domain extension appears to the right of the punctuation mark, while the domain name appears to the left of the punctuation mark. Additionally, top level domains (TLD), also known as first-level domains, and sub level domains can be used to categorize the domain structure (SLD).

The most common TLDs include .com, .info, .org and .net. A top-level domain is also sometimes represented as a country code such as .us, .ca and .lk. Sub level domains are domains that are created in order to connect a related page to the main website.

Sub-level domains are parts of a top-level domain that represent a particular good or service, group of people, or company. They avoid the need to register an entirely other domain and point users to a specific section of an existing website. The fact that domain names are distinctive is one of their best qualities. Some domain names are highly valuable because of their uniqueness. (Morgan Joel, EDUCBA, 2022)

Review the effect of search engines on website performance. Provide evidence-based support for improving a site's index value and rank through search engine optimization.

Typically, a search engine is an online software program or script that looks for keywords in documents and files and returns the results of any files that include those keywords. The Internet is now home to a wide variety of search engines, each having unique capabilities and features as well as being quite potent. However, Veronica, the first text-based search engine, is regarded as the first search engine ever created. Archie was used to search for file transfer protocol (FTP) files. Whatever, Google is currently the most well-liked, effective, and search engine.

Effect of search engines on website performance is crucial since it affects how well a website interacts with users. It benefits users by making it simple for them to access the website. Another crucial benefit is that it improves the website's performance because people are less likely to find a website that does not appear in the top results or on the first page of the search result section. The website might not be correctly found by any respectable search engines, which would produce unfavorable results. However, there are also very significant effects of search engines on the performance of websites.

Search engines download information from websites and store it in databases as part of the website crawling process. They do this because they need to process data so that their users may see the most pertinent results. Therefore, web page indexing is

crucial for websites to be presented in search engines in response to user queries, and search engines offer a variety of tools to aid in enhancing indexing of any website.

Here are the techniques. The first one is an XML sitemap, or XML (Extensible Markup Language) sitemap, which is a collection of webpage URLs that websites use to help search engines understand how they want to be crawled. The second one is a Meta robots tag, which tells search engines to look for it whenever they visit a webpage.

This tag tells them whether or not they are permitted to index this page. The last one is robots.txt, which is a strong way to enable or restrict search engines from massively indexing websites and is situated in the root directory of the website. For instance, the first step is to include a meta no index tag throughout the entire website if we don't want any search engines to scan it. If the website is not dynamic, that can be a little frantic and there's a risk you'll skip certain webpages. These are the techniques for indexing websites.

Search engines are a typical starting place when you require information, whether you're looking into a particular product, finding a restaurant, or making travel arrangements. They provide an excellent chance for business owners to send targeted visitors to your website.

The process of positioning your website to appear higher on a search engine results page (SERP) in order to draw in more traffic is known as search engine optimization (SEO). For search terms that are most important to your target audience, it is common to try to rank on the first page of Google results. Therefore, understanding your audience's wants and needs is just as important to SEO as knowing the technical details of how to set up your website. (Mia archeat, mailchimp, 2021)

1.3

Identify and explain the common web development technologies and frameworks.

Every website developer should be familiar with the extensive set of rules and procedures that come with web development. You need to be knowledgeable about web technologies that will help you accomplish your goal if you want a website to look and function the way you want it to. Frameworks have become a significant component of the online development industry as a result of the expanding standards for web applications and, consequently, the rising complexity of technology. It might make more sense to endorse in frameworks rather than building new methods. Frameworks aid programmers in creating interactive and sophisticated apps.

Definition of Web development

Web development refers to the creating, building, and maintaining of websites. It includes aspects such as web design, web publishing, web programming, and database management. It is the creation of an application that works over the internet i.e. websites.

Computers need codes to communicate with one another because they can't speak to each other like people can. The markup languages and multimedia applications that computers use to communicate are known as web technologies. (Jon brahch, Computerhope, 2017)

Common Web technologies

1. **Browsers**

Alternatively referred to as a web browser or Internet browser, a browser is a software program to present and explore content on the World Wide Web. These pieces of content, including pictures, videos, and web pages, are connected using hyperlinks and classified with URIs (Uniform Resource Identifiers). This page is an example of a web page that can be viewed using a browser.

There have been many different web browsers that have come and gone over the years. The first, named WorldWideWeb (later changed to Nexus), was invented by Tim Berners-Lee in 1990. However, the first graphical browser and widely used browser that help bring popularity to the Internet was NCSA Mosaic.

For ex:-

Google Chrome – Currently, the most popular browser brought to you by
Google Safari – Apple's web browser

Firefox – Open-source browser supported by the Mozilla Foundation (Jon brahch, Computerhope, 2017)

2. **HTML & CSS**

One of the first things you should learn is HTML. HTML helps web browsers understand what to display whenever they receive a request. You must be familiar with CSS if you want to comprehend HTML more fully. Cascading Style Sheets, or CSS, specify how HTML elements should appear on the screen. You can quickly develop CSS text effects, page transitions, picture hover effects, and more if you look through enough tutorials. (Jon brahch, Computerhope, 2017)

3. **Programming Languages**

Using programming languages, you may instruct computers on what to do. Similar to how there are numerous linguistic languages, there are numerous different computer languages (English, Spanish, French, Chinese, etc.). The

one is not superior to the other. Developers often excel at a small number of things, so they prioritize promoting those. Some of the languages and links to their home sites are included below.

- **JavaScript** : used by all web browsers, Meteor, and lots of other frameworks
- **Python** :used by the Django framework and used in a lot of mathematical calculations
- **PHP** : used by WordPress
- **Objective-C**: the programming language behind iOS (your iPhone), led by Apple
- **Swift** :Apple's newest programming language
- **Java**: Used by Android (Google) and a lot of desktop applications. (Jon brahch, Computerhope, 2017)

Frameworks

"Framework" is a word used a lot in relation to code. So, what even is a framework? When we talk about frameworks, those frameworks normally fall into three different categories -- frontend, backend, and CSS or user interface frameworks. But even once you know what frameworks are, which one should you learn or use and why? This post will discuss what a framework is, what frameworks are out there, and how to make decisions about what kind of framework you should use for your projects!

If we look at a couple different web applications we can see that there are a lot of duplicated features. Many have user authentication, pages that render, they connect to databases, they have profiles you can look at, they all have stylized feeds of information etc. If I wanted to, I could write all of that logic myself every time I want to build a new application; however, there are also a lot of tools available which can do that for us instead.

In addition, a lot of the time when we build applications we want to format our code in certain ways that help us become more efficient and help other developers understand our code. Web frameworks help us achieve structure in our applications, and they give us additional features we can add to them without too much extra work. Frameworks give us a place to start so that we can focus on features rather than configuration details.

Code is read many more times than it is written, and it is so important when you are working with other developers that they know where the code for certain tasks is being written so they can use it in their own code. The frameworks also have conventions for how the code is written and structured, which standardize how the developers write their code. (Bobby brycedale, Welearncode, 2022)

Common web frameworks

1. Express

Express is currently the most popular backend framework, which can be attributed to the surge in Node.js's popularity. Its compatibility with other frameworks like Kraken, loopback, and sails makes it a favorite among heavyweight competitors like Uber, IBM, and Accenture. It has a simple, quick, and non-opinionated framework.

The fundamental features it offers to make use of Node.js performance without hiding its capabilities. It supports complete and REST API applications due to its versatility. There is only one problem with it: there is no clear way to go about things, so newcomers could feel a little bewildered. (Bobby brycedale, Welearncode, 2022)

2. Django

Django is a Python-based model-view-template framework for building websites. Large corporations like Google, YouTube, and Instagram as well as website development firms make use of it. Its most notable features are some battery-included functions like messaging and authentication. It adheres to the convention over configuration and DRY patterns.

Django views security as somewhat vital and offers developers methods and tools to aid in creating secure websites and putting security measures into place throughout the framework. Preventing code execution within the template layer is one instance. (Bobby brycedale, Welearncode, 2022)

3. Rails

Rails is a model-view-controller framework for Ruby that is favored by many developers. GitHub, Hulu, Shopify, Airbnb, and GitHub are some of the major Rails users. Rails is regarded as a beginner-friendly framework and has a ton of helpful gems. With its library-like dependencies that reach the functionalities of programs, developers may work more quickly and effectively. There are a ton of tutorials, screencasts, and resources online if you're interested in learning more about this technology. The Rails community is amiable and somewhat trustworthy. Rails' only drawback is how long it takes to deploy and execute in production. And once you delve in, the training curve is sort of steep. (Bobby brycedale, Welearncode, 2022)

4. Laravel

Laravel is a PHP-based model-view-controller framework. The most used web language is PHP. This framework is relatively new and includes built-in API support. Because it has a respectable number of packages, its range is frequently increased. From its screenshot instructional website and more than a thousand videos on PHP, Laravel, and front-end technologies, beginners may learn a ton.

Developers can build specialized infrastructure and back out common tasks like authentication, sessions, storage, and steering thanks to Laravel's beautiful

and expressive structure. Laravel emerges as the top choice of many developers and website development businesses due to its scalability, performance, and functionality. (Bobby brycedale, Welearncode, 2022)

5. Spring

Another Java-based Model-View-Controller framework is this one. Websites like Wix, Ticketmaster, and BillGuard use spring. Java is well-liked by developers since it is a tightly typed language.

It offers enterprise-grade app development with the main goal of streamlining the creation of J2EE apps. With its strict architecture structure, developers may produce straightforward, quick, and adaptable apps. (Bobby brycedale, Welearncode, 2022)

Task 2

2.1

Define the relationships between front-end and back-end website technologies and discuss how the front-end and the back-end relate to presentation and application layers.

Front End Development

The part of a website that the user interacts with directly is termed the front end. It is also referred to as the 'client side of the application. It includes everything that users experience directly: text colors and styles, images, graphs and tables, buttons, colors, and a navigation menu. HTML, CSS, and JavaScript are the languages used for Front End development. Responsiveness and performance are the two main objectives of the Front End. The developer must ensure that the site is responsive i.e. it appears correctly on devices of all sizes no part of the website should behave abnormally irrespective of the size of the screen.(Palaksingh,greeksforgeeks,2020)

Back End Development

The backend is the server side of the website. It stores and arranges data, and also makes sure everything on the client side of the website works fine. It is the part of the website that you cannot see and interact with. It is the portion of software that does not come in direct contact with the users. The parts and characteristics developed by backend designers are indirectly accessed by users through a front-end application. Activities, like writing APIs, creating libraries, and working with system components without user interfaces or even systems of scientific programming, are also included in the backend. (Palaksingh, greeksforgeeks, 2020)

In web applications, the term front end refers to the code performed in the browser, and the back end to code being executed on the web server. The front end is commonly termed the client side while the back end is the server side. JavaScript code that uses HTML for presentation and CSS for styling can be run by browsers. The back-end is significantly more flexible and can be programmed in many different languages, with PHP, Ruby, ASP.NET, and JavaScript being the most popular ones (Node). The application, the database, and the back-end development will all be used to store the data required to host a website on servers.

A website's database is crucial since it saves data and is required for website development. MySQL 8 is a popular database that is utilized by many of the well-known global brands and corporations, like Google, Facebook, and PayPal. In order for MySQL to function, a database must first be created for storing and modifying data. Clients can then submit requests by typing specific SQL queries into MySQL, and the server's application will reply by displaying the desired data on the client's browser.

The back-end development of a website will have the language, database, security and backup side of it covered. The database of a website will be used in a query when someone is for example searching for hotel within a certain town, the database is responsible for accepting the query made by someone then it will find the data requested in the database and then return it to the user that has searched for it on a website.

Data and sensitive information must be kept safe, hence the back-end development's security component is essential. When performing any development, it is crucial that users log in via the HTTPS protocol because this will encrypt the data and prevent hackers from intercepting it. Additionally, it is crucial that websites that accept credit card information use HTTPS as this will protect the data from hackers and give users confidence to use websites that adhere to this protocol. For instance, 84% of users would abandon a purchase if they cannot tell that a website is utilizing the HTTPS protocol. Google will rank sites higher in their search results if a site uses the HTTPS protocol, so it is better to use it in terms of getting your website higher up in the search results.

The front end of a website is what you see; for instance, a browser will display the results of a search query. HTML is a part of front-end development, and when

combined with CSS and JavaScript, it may be used to create a website's front end. These coding standards are used to generate the website pages, which the developer then styles according to the preferences of the customer. A web developer is the individual or group who creates the pages of a website that we see on the screen. They are able to construct static websites, which are websites without any back-end code or databases. These static websites are used to display data for a website that does not require continual updating, such as a restaurant that may have one to display only the menus and location. Since the website might not require updated for a while, they don't need the data to be kept in a database.

The presentation layer in the OSI model is responsible for translating data, encryption between applications and data compression. When data is passed through the HTTPS protocol on a browser, the data is encrypted as to keep it secure from people trying to hack and intercept it when it is transmitted over the internet. When this occurs the two devices that are talking to each other, for example when someone buys something off the internet, the HTTPS protocol is used to keep credit card information safe. When this happens the buyer and the seller will exchange keys to encrypt the data that is being transmitted, and this means that only these two can encode the data and then decode it in to a readable format, to keep it secure even if it does get intercepted by a hacker, they cannot retrieve any information from the hacked data as they do not have a key to decode it.

When uploading files to a web server to host a website, for example, the FTP protocol is used in the application layer to transmit files between the server and the client over a network. The application layer, which is the sole layer that interacts with computer users, offers the interface between applications and the network. There are numerous other protocols used at the application layer, including SMTP, Telnet, and TFTP. The DNS protocol is also used here. This layer is used by apps that operate on computers because it hosts the programs that those programs employ to access the internet. The mail protocol, which is part of the application layer, employs the SMTP protocol to send emails over the internet, and the POP3 protocol to receive them. The Telnet protocol, which is used to access a device remotely, is also on the application layer. (Palaksingh, greeksforgeeks, 2020)

2.2

Critically compare the different between online website creation tools and custom-built sites considering the Followings: design flexibility, performance, functionality, User Experience (UX) and User Interface (UI). Evaluate the tools and techniques available to design and develop a custom-built web Applications.

Capabilities and relationships between front-end and back-end website technologies

Technologies used by front end developers

While there are many different types of technologies and stacks, most front-end web developers use HTML, CSS and JavaScript, the de facto building blocks of the web, and client-side frameworks such as Angular, React, Stencil and Vue.

Not everything happens on the frontend, though. Client-side rendered applications still rely on services and APIs that run on back-end remote servers or the cloud. (Pluralsight, 2023)

What kind of jobs do front end developers have?

- **Web designer:** A web designer, you guessed it, designs websites. The job title of web designer is pretty broad, though. A web designer could just be someone who designs the sites in a program like Photoshop or Fireworks and will never touch the code. But in another location, a web designer could do all the design comps in Photoshop and then be responsible for creating all the HTML and CSS (and sometimes even JavaScript) to go along with it.
- **User interface (UI) designer:** This is basically a visual designer and is generally focused on design. They're not usually involved in the implementation of the design, but they might know light HTML and CSS so they can communicate their ideas more effectively to the front-end developers.
- **User experience (UX) designers:** UX designers work in the frontend, studying and researching how people use the sites. Then, they make changes through a lot of testing.
- **Front-end developer:** Also called a front-end designer, they can create a site without any back-end development. The site they would create without a web developer, or using the backend, is a static site. A static site is something like a site for a restaurant or hair salon. It doesn't require any information to be stored in a database. The pages will almost always stay the same, unless it's time for a redesign. A front-end developer may be required to have a grasp on testing, as well as be well versed in HTML, CSS and JavaScript. This person may or may not have experience with creating the design in a design program. A different version of this title is front-end engineer. People who work with specific front-end languages like JavaScript developer are also considered front-end developers. (Pluralsight, 2023)

Technologies used by backend developers

When back-end developers create apps that render on the server-side, they use the same building blocks as front-end developers: HTML, CSS and JavaScript.

Back-end developers also work with software stacks that include operating systems, web servers, frameworks, languages, programming APIs and more. The frameworks,

languages and programming APIs in these stacks are used to render server-side sites and web applications and to create services that other applications can consume.

Notorious stacks include .NET, MEAN and LAMP, but there are many more, and each includes a programming language of choice, such as C#, JavaScript, Java, Go, Python or PHP. (Pluralsight, 2023)

What kind of jobs do back end developers have?

The behind-the-scenes functionality of a website is referred to as the 'back-end'. It is a combination of technology and programming that powers a website, consisting of three parts that a user never sees: a server, an application, and a database.

The back-end runs a website — the user doesn't see it or interact with it, but it is always running in the background, delivering functionality and a desktop-like experience. Back-end developers play a critical role in web development teams, and make sure that data or services requested by the front-end system or software are delivered.

The back end is the combination of a database and software written in a server-side language, which is run on web servers, cloud-based servers, or a hybrid of both. In order to make the server, application, and database communicate with each other, back-end developers use languages like PHP, Ruby, Python, Java, and .Net to build an application, and tools like MySQL, Oracle, and SQL Server to find, save, or change data and serve it to the user in front-end code.

After communicating with a business owner and understanding what their needs and technical requirements are, back-end developers use the aforementioned tools to create web applications with clean, well-documented code. They come up with the most effective and efficient solutions for engineering technology by converting and returning data into front-end code a user can interact with, such as: filling out a form, creating a profile, shopping online, etc.

Responsibilities of a back-end developer can include working with:

- Database creation, integration, and management
- Back-end frameworks to build server-side software
- Web server technologies
- Cloud computing integration
- Server-side programming languages
- Operating systems

- Content management system development, deployment, and maintenance
- API integration
- Security settings and hack prevents
- Reporting — generating analytics and statistics
- Backing up and restoring technologies for a website's files and DB

(Pluralsight, 2023)

